

Part XI Dermatology - Chapter 11E (Revised)

Skin Cancer, Premalignant Lesions, and Pigmented Lesions

Expanded numeric guideline version with excision margins, Breslow thresholds, biopsy strategy, Mohs logic, follow-up, and red flags

This regenerated chapter expands the skin cancer segment with more numbers and guideline-style thresholds. It is written for internal medicine learners who need to recognize dangerous lesions, choose biopsy/referral rather than empiric treatment, understand pathology language, and interpret surgical plans. It keeps the original benign and premalignant lesion coverage from the uploaded dermatology source - actinic keratosis, seborrheic keratosis, and keratoacanthoma - but adds basal cell carcinoma, cutaneous squamous cell carcinoma, melanoma, Merkel cell carcinoma, Kaposi sarcoma, cutaneous T-cell lymphoma, surveillance, and prevention.

1. First-pass algorithm for a suspicious lesion

Begin with change, symptoms, morphology, site, and host risk. A lesion can be malignant even when small. Many early melanomas are less than 6 mm, and early BCC or SCC can mimic eczema, acne, trauma, or a wart. Do not repeatedly freeze or steroid-treat a lesion that is evolving, bleeding, ulcerated, painful, indurated, or nonhealing.

Step	Question	Why it matters
1	Changing, bleeding, ulcerating, painful, nonhealing, asymmetric?	These features override a reassuring first impression and usually warrant biopsy/referral.
2	Pigmented? Apply ABCDE plus ugly duckling.	Evolution and discordance from other nevi are especially important.
3	Rough or scaly on sun-exposed skin?	Actinic keratosis, SCC in situ, and invasive SCC are common possibilities.
4	Pearly, translucent, telangiectatic, rolled, scar-like, or slowly ulcerating?	Classic BCC patterns; infiltrative BCC may look like a scar.
5	Immunosuppressed, transplant, CLL, HIV, chronic biologics/steroids?	SCC is more frequent, recurrent, and aggressive.
6	H-zone face, ear, lip, genitalia, hand, foot, nail unit, scar, chronic ulcer, radiation field?	High-risk site can change treatment from simple excision to Mohs/staged or specialty management.

Red flag: biopsy or refer any lesion that is evolving, nonhealing beyond about 4 to 6 weeks without a clear benign explanation, spontaneously bleeding, ulcerated, rapidly enlarging, painful, neurologically symptomatic, or located in a high-risk site.

2. Documentation and bedside description

Record lesion type, exact location with laterality, largest diameter in millimeters, color, surface, border, symptoms, duration, growth rate, prior treatment, and whether a photograph was taken. For pigmented lesions, document ABCDE and ugly-duckling status. For keratinocyte lesions, document scale, crust, keratin horn, ulceration, tenderness, induration, fixation, and regional lymph nodes if high-risk SCC is possible.

Descriptor	Examples	Implication
Size	Largest diameter in mm; melanoma also needs Breslow depth from pathology.	Surface diameter helps BCC/SCC risk; Breslow depth drives melanoma staging.
Surface	Scale, crust, hyperkeratosis, rolled edge, telangiectasia, central keratin plug.	Scale/keratin suggests AK/SCC; rolled translucent edge suggests BCC.
Distribution	Sun-exposed, acral, nail, mucosal, scar, chronic wound, radiation field.	Acral/nail/mucosal melanoma may lack classic sun history.
Symptoms	Pain, itch, bleeding, numbness, tingling.	Pain/paresthesia can suggest perineural involvement.
Host	Transplant, CLL, HIV, chronic immunosuppression, tanning bed, prior radiation.	Lower threshold for biopsy and closer surveillance.

3. Actinic keratosis and field cancerization

Actinic keratoses are premalignant keratinocyte proliferations caused by cumulative ultraviolet damage. They are often more palpable than visible: gritty, erythematous or skin-colored scaly macules or papules on scalp, face, ears, dorsal hands, and forearms. The uploaded source describes AKs as small rough scaly lesions due to repeated sun exposure and emphasizes biopsy for hypertrophic AKs or lesions that become indurated, tender, or bleed spontaneously to exclude SCC.

Finding	Simple AK pattern	Concerning for SCC/SCC in situ
Thickness	Thin gritty macule or papule.	Thick hyperkeratotic plaque, cutaneous horn, indurated base.
Symptoms	Asymptomatic or mildly tender.	Painful, rapidly enlarging, bleeding, ulcerated.
Response	Improves after cryotherapy or field treatment.	Persists or recurs quickly after appropriate treatment.
Management	Cryotherapy for isolated lesions; field therapy when many lesions.	Biopsy before repeated destructive therapy.

Numeric anchors: topical 5-fluorouracil field treatment is often prescribed for about 2 to 4 weeks depending on formulation and site. Isolated AKs can be treated with cryotherapy; broad fields can be treated with 5-FU, imiquimod, diclofenac, tirbanibulin, or photodynamic therapy. The transformation risk of any single AK is low, but multiple AKs identify a UV-damaged field with elevated patient-level SCC risk.

4. Seborrheic keratosis and benign mimics

Seborrheic keratoses are benign epidermal tumors, usually appearing after age 30, often waxy, verrucous, tan-brown to black, and stuck-on. The uploaded source notes that they are common, hereditary, harmless, and have no malignant potential. The major clinical pitfall is that melanoma, pigmented BCC, and SCC in situ can mimic an irritated seborrheic keratosis.

Lesion	Classic clue	When to biopsy
Seborrheic keratosis	Waxy stuck-on cerebriform papule/plaque; often multiple similar lesions.	New solitary dark lesion, irregular pigment, spontaneous bleeding, rapid change, or ugly duckling.
Dermatofibroma	Firm papule, often leg, dimples with lateral compression.	Rapid growth, ulceration, atypical color, or diagnostic uncertainty.
Cherry angioma	Bright red dome papule, often multiple on trunk.	Bleeding pigmented vascular lesion or concern for amelanotic melanoma.
Pyogenic granuloma	Friable rapidly growing red papule that bleeds easily.	Adult new rapidly growing lesion; amelanotic melanoma/SCC can mimic.

5. Basal cell carcinoma

Basal cell carcinoma is the most common human malignancy. It grows slowly and rarely metastasizes, but it is locally destructive. Nodular BCC is classically pearly with arborizing telangiectasias, rolled borders, and central ulceration. Superficial BCC can mimic eczema or psoriasis as a pink scaly patch. Morpheaform or infiltrative BCC may appear as a scar-like indurated plaque with subclinical extension.

BCC feature	Low-risk pattern	High-risk pattern
Location/size	Small, well-defined trunk or extremity lesion.	H-zone face, eyelid, nose, ear, lip, temple, genitalia, hands/feet, or large lesion.
Borders	Well-defined clinical edge.	Poorly defined, recurrent, scar-like, infiltrative.
Histology	Nodular or superficial.	Morpheaform, infiltrative, micronodular, basosquamous, perineural invasion.
Host	Immunocompetent, no prior radiation.	Immunosuppression, prior radiation, genetic syndromes, multiple recurrent tumors.

Number to know: low-risk primary BCC treated by standard excision commonly uses a 4-mm clinical margin with histologic margin assessment. High-risk BCC is often managed with Mohs micrographic surgery, staged excision, or other specialist-directed treatment rather than a casual wider excision.

Treatment	Typical use	Numbers/practical details
Standard excision	Low-risk primary BCC when closure feasible.	Common margin: 4 mm for low-risk primary BCC.
Mohs micrographic surgery	High-risk site, recurrent tumor, aggressive histology, tissue-sparing need.	Complete peripheral/deep margin assessment; favored for H-zone face and recurrent/aggressive tumors.
Curettage/electrodesiccation	Selected low-risk superficial/nodular lesions on trunk/extremities.	Avoid when margin control matters or high-risk features exist.
Topical therapy	Selected superficial low-risk BCC.	Not for nodular high-risk, infiltrative, recurrent, or deep tumors.

6. Cutaneous squamous cell carcinoma and SCC in situ

Cutaneous SCC ranges from SCC in situ (Bowen disease) to invasive tumors with metastatic potential. It often presents as a firm hyperkeratotic papule, plaque, nodule, cutaneous horn, or nonhealing ulcer on sun-exposed skin. SCC may be tender. Compared with BCC, SCC has greater metastatic potential, especially in immunosuppressed patients and high-risk sites such as lip and ear.

High-risk SCC feature	Threshold or clue	Why it matters
Diameter	Often ≥ 2 cm on trunk/extremities; smaller sites can still be high-risk.	Larger tumors recur/metastasize more.
Depth	Deep invasion, beyond subcutaneous fat, fixed/deep lesion.	May require imaging, Mohs, radiation, oncology.
Histology	Poor differentiation, desmoplastic, perineural invasion.	Higher recurrence/metastasis.
Symptoms	Pain, numbness, tingling, weakness.	May suggest perineural invasion.
Host/site	Transplant, CLL, HIV; ear, lip, genitalia, nail, scar, radiation field.	More aggressive behavior; lower biopsy threshold.

Number to know: low-risk primary cutaneous SCC treated by standard excision commonly uses 4- to 6-mm clinical margins. High-risk SCC often needs Mohs, staged excision, nodal assessment, imaging, radiation, or multidisciplinary care.

Entity	Clue	Management anchor
SCC in situ	Persistent erythematous scaly plaque; can mimic eczema.	Biopsy; excision, ED&C, topical 5-FU/imiquimod, PDT, or Mohs depending site/risk.
Invasive SCC	Tender keratotic papule/nodule, ulcer, cutaneous horn.	Biopsy deep enough for invasion; treat based on risk.
Keratoacanthoma-type lesion	Rapid crateriform dome with central keratin plug over weeks.	Often managed as SCC-spectrum lesion; excise/refer rather than observe if uncertain.
Marjolin ulcer	SCC in chronic wound, burn scar, sinus tract.	Aggressive; biopsy chronic nonhealing wounds.

7. Melanoma recognition and biopsy

Melanoma causes disproportionate skin cancer mortality. The internist's key role is not dermoscopic certainty but knowing when to biopsy or refer promptly. A suspicious melanoma should usually undergo complete excisional biopsy with narrow margins, deep enough to include the full lesion and subcutaneous fat if possible. Partial biopsy may under-stage Breslow depth, but it may be necessary for large facial, acral, mucosal, or nail lesions; sample the thickest/darkest/most atypical area.

Tool	Details	Clinical use
ABCDE	Asymmetry, Border irregularity, Color variation, Diameter often >6 mm, Evolution.	Evolution is the most important history clue; small melanomas occur.
Ugly duckling	Looks different from patient's other nevi.	Useful when many atypical nevi are present.
EFG	Elevated, Firm, Growing.	Helps identify nodular melanoma, which may be symmetric.
Acral/nail clues	Irregular acral pigmentation, widening melanonychia, Hutchinson sign.	Do not reassure based on lack of sun exposure.

Biopsy technique	Preferred use	Numeric/practical detail
Excisional biopsy	Most suspicious pigmented lesions when feasible.	Narrow 1- to 3-mm clinical margins; full thickness to deep dermis/subcutis.
Deep saucerization/shave	Raised lesions or sites where ellipse impractical.	Must be deep enough to avoid transecting melanoma base.
Punch/incisional biopsy	Large facial, acral, or nail lesions.	Sample most atypical/thickest area; risk of under-staging.
Avoid superficial shave	Possible melanoma.	Can transect lesion and compromise Breslow measurement.

8. Melanoma pathology and margin numbers

The pathology report must include Breslow thickness in millimeters, ulceration, mitotic activity, margin status, microsatellitosis, lymphovascular invasion, neurotropism/perineural invasion, and subtype. Breslow thickness is measured from the granular layer, or from the base of an ulcer, to the deepest invasive melanoma cell. It drives T category, wide excision margin, and sentinel lymph node biopsy discussions.

T category concept	Breslow depth	Ulceration modifier
Tis	Melanoma in situ, epidermis only.	No dermal invasion.
T1	≤1.0 mm	T1a generally <0.8 mm nonulcerated; T1b includes 0.8-1.0 mm or ulceration.
T2	>1.0-2.0 mm	a = no ulceration; b = ulcerated.
T3	>2.0-4.0 mm	a = no ulceration; b = ulcerated.
T4	>4.0 mm	a = no ulceration; b = ulcerated.

Melanoma after diagnostic biopsy	Common clinical excision margin
Melanoma in situ	0.5 to 1.0 cm; lentigo maligna/head-neck lesions may need staged excision or Mohs-type margin control.
Invasive melanoma ≤1.0 mm	1 cm.
Invasive melanoma >1.0 to 2.0 mm	1 to 2 cm, individualized by site and closure.
Invasive melanoma >2.0 mm	2 cm when anatomically feasible.

Sentinel lymph node biopsy threshold: discuss SLNB for melanoma >0.8 mm, thin melanoma <0.8 mm with ulceration or adverse features, and generally for melanoma >1.0 mm if clinically node-negative. Palpable nodes require diagnostic imaging/biopsy rather than SLNB.

Scenario	SLNB discussion
Melanoma in situ	No SLNB.
<0.8 mm and nonulcerated	Usually no SLNB unless exceptional adverse features.
<0.8 mm ulcerated or 0.8-1.0 mm	Discuss SLNB.
>1.0 mm	Generally offered/discussed if clinically node-negative.

Scenario	SLNB discussion
Palpable node	Image and biopsy node; do not use SLNB as next step.

9. Melanoma subtypes and pitfalls

Subtype	Pattern	Pitfall
Superficial spreading	Most common; irregular patch/plaque with variable pigment.	Many arise de novo, not only from moles.
Nodular	Elevated, firm, growing papule/nodule; may be black, blue, pink, or amelanotic.	May not meet ABCD; use EFG.
Lentigo maligna	Sun-damaged face/scalp/neck of older adults.	Subclinical extension; staged excision often useful.
Acral lentiginous	Palms, soles, nail unit.	Often diagnosed late; biopsy changing acral/nail pigment.
Amelanotic	Pink/red papule, plaque, or nodule.	Mimics pyogenic granuloma, BCC, SCC, dermatitis.

10. Other dangerous tumors

Merkel cell carcinoma, Kaposi sarcoma, and cutaneous T-cell lymphoma are less common but important. Merkel cell carcinoma is aggressive and often presents as a rapidly expanding, firm, painless red-violaceous or skin-colored nodule in an older or immunosuppressed patient. Kaposi sarcoma is HHV-8-associated and appears as violaceous macules, plaques, or nodules, especially with HIV/AIDS or transplant immunosuppression. Mycosis fungoides can look like chronic eczema or psoriasis in bathing-trunk distribution for years.

Disease	Clinical clue	Triage
Merkel cell carcinoma	AEIOU: Asymptomatic, Expanding rapidly, Immune suppression, Older than 50, UV-exposed fair skin.	Urgent biopsy/referral; staging and nodal evaluation.
Kaposi sarcoma	Violaceous macules/plaques/nodules; oral, leg, genital, widespread.	Evaluate HIV/immune status; biopsy uncertain lesions.
Cutaneous T-cell lymphoma	Persistent patches/plaques, poikiloderma, pruritus, bathing-trunk distribution.	Biopsy persistent/refractory lesions; repeat biopsies may be needed.

11. Mohs logic and high-risk sites

Mohs micrographic surgery is tissue-sparing complete margin assessment, not simply a larger excision. It is valuable when recurrence risk is high, subclinical extension is likely, or tissue preservation matters. Location alone can make a lesion high-risk.

Zone/site	Examples	Reason
H-zone face	Central face, nose, eyelids, lips, chin, mandible, peri/pre/postauricular skin, temple, ear.	Subclinical spread, recurrence risk, cosmetic and functional preservation.
Moderate-risk sites	Cheeks, forehead, scalp, neck, jawline.	Risk depends on size, histology, recurrence, borders, host.
Special sites	Hands, feet, nail unit, genitalia, nipples/areola.	Function and complex closure; metastatic implications for SCC.
Scar/radiation/chronic wound	Burn scars, chronic ulcers, sinus tracts, irradiated fields.	SCC may be aggressive; biopsy chronic nonhealing wounds.

12. Immunosuppression and skin cancer

Immunosuppression shifts risk toward multiplicity, recurrence, and aggressive SCC. Solid organ transplant recipients, CLL patients, advanced HIV patients, and patients on chronic immunosuppressive therapies deserve lower biopsy thresholds and more frequent full-skin surveillance.

Group	Concern	Management
Solid organ transplant	Markedly increased SCC risk; tumors can be multiple and aggressive.	Regular dermatology exams, sun protection, early biopsy, coordinate immunosuppression if severe burden.
CLL/hematologic malignancy	Aggressive keratinocyte cancers; immune dysfunction.	Lower referral and nodal evaluation threshold.
HIV/AIDS	Kaposi sarcoma, HPV tumors, extensive molluscum/warts, SCC.	Assess immune status and ART; biopsy atypical lesions.
Biologics/JAK inhibitors/steroids	Variable NMSC risk by drug/disease.	Document skin history; counsel and surveil.

13. Follow-up and prevention numbers

Condition after treatment	Typical follow-up framework	Notes
AK/field cancerization	Often every 6-12 months depending burden.	Biopsy thick/tender/persistent lesions.

Condition after treatment	Typical follow-up framework	Notes
Low-risk BCC	Periodic skin exams; often 6-12 months initially, then annually if stable.	High risk for additional BCCs.
SCC	At least annual; high-risk SCC often every 3-6 months initially.	Assess regional nodes in high-risk disease.
Melanoma in situ/stage I-II	Often every 3-12 months for first 5 years depending stage, then annually.	Self-skin and lymph node awareness.
Higher-stage melanoma	More frequent and oncology-directed; imaging as indicated.	Coordinate recurrence and immunotherapy toxicity monitoring.
Transplant/highly immunosuppressed	Often every 3-12 months depending tumor burden.	May need field therapy or chemoprevention by specialists.

- Use broad-spectrum sunscreen SPF 30 or higher and reapply about every 2 hours outdoors or after swimming/sweating.
- Avoid indoor tanning. Counsel protective clothing, wide-brim hats, UV-blocking sunglasses, and shade, especially midday.
- Teach monthly self-skin checks for melanoma history, many atypical nevi, transplant status, or multiple prior skin cancers.

14. Integrated diagnosis and treatment traps

Presentation	Likely considerations	Next step
Gritty scaly macules on bald scalp/face/dorsal hands	Actinic keratoses/field cancerization.	Cryotherapy or field therapy; biopsy thick/tender/bleeding lesions.
Pearly papule with telangiectasia/rolled border	Nodular BCC.	Biopsy; excision or Mohs depending risk.
Pink scaly plaque persistent despite steroid	SCC in situ, superficial BCC, inflammatory mimic.	Biopsy, not repeated steroid.
Tender keratotic nodule or cutaneous horn	Invasive SCC until proven otherwise.	Deep biopsy and risk-based treatment.
Rapid crateriform keratin-filled nodule	Keratoacanthoma-type SCC.	Excise/refer.
Changing irregular pigmented lesion	Melanoma, atypical nevus, pigmented BCC/SK.	Excisional biopsy with 1-3 mm margins when feasible.

- Do not freeze a pigmented lesion unless diagnosis is secure; cryotherapy destroys tissue and may delay melanoma diagnosis.
- A lesion that returns after cryotherapy deserves biopsy.
- BCC margin number: 4 mm for low-risk primary BCC. Low-risk cSCC margin number: 4-6 mm. Melanoma margins are in centimeters and depend on Breslow depth.
- Breslow thickness is more important than surface diameter for melanoma staging. A small nodular melanoma can be thick and dangerous.
- Seborrheic keratosis is benign, but melanoma can mimic an irritated SK; biopsy the ugly duckling or changing lesion.

15. Compact numeric summary

Decision point	Number/threshold	Use
Suspicious melanoma biopsy margin	1-3 mm	Diagnostic excisional biopsy when feasible.
Melanoma in situ excision	0.5-1.0 cm	Definitive excision or staged control depending site/subtype.
Melanoma ≤1.0 mm	1 cm	Post-biopsy wide excision.
Melanoma >1.0-2.0 mm	1-2 cm	Post-biopsy wide excision, individualized.
Melanoma >2.0 mm	2 cm	Post-biopsy wide excision if feasible.
SLNB discussion	>0.8 mm or ulcerated/adverse thin melanoma; generally >1 mm	Staging and prognosis if clinically node-negative.
Low-risk BCC excision	4 mm	Standard excision with histologic margin assessment.
Low-risk cSCC excision	4-6 mm	Standard excision with histologic margin assessment.
Nonhealing lesion	Persistent about 4-6 weeks	Biopsy/referral.
Sunscreen	SPF ≥30; reapply every 2 hours outdoors	Prevention counseling.

16. Graph-RAG concept map seeds

Core nodes: actinic keratosis, field cancerization, SCC in situ, invasive SCC, basal cell carcinoma, melanoma, Breslow thickness, sentinel lymph node biopsy, Mohs micrographic surgery, keratoacanthoma, seborrheic keratosis, dysplastic nevus, Merkel cell carcinoma, Kaposi sarcoma, cutaneous T-cell lymphoma, immunosuppression, transplant-associated SCC, UV radiation, tanning bed exposure, perineural invasion, H-zone face, ABCDE criteria, ugly duckling sign.

Suggested edges: UV radiation causes DNA damage leading to AK/BCC/SCC/melanoma; AK is a premalignant precursor and risk marker for SCC; immunosuppression increases SCC aggressiveness; Breslow thickness determines melanoma T category and excision margin;

ulceration worsens melanoma prognosis; SLNB stages clinically node-negative melanoma above threshold; Mohs treats high-risk keratinocyte carcinoma by complete margin assessment; seborrheic keratosis mimics melanoma but has no malignant potential; keratoacanthoma resembles SCC and is often managed as SCC-spectrum disease when uncertain.

Selected references and source guide

- Uploaded dermatology packet: actinic keratosis, seborrheic keratosis, keratoacanthoma, fungal/nail context, and lesion-recognition material; used as local coverage guide and synthesized into original text.
- American Academy of Dermatology guideline highlights for melanoma, basal cell carcinoma, and cutaneous squamous cell carcinoma.
- NCCN-derived surgical margin frameworks as summarized in AAD quality measures, NCI PDQ, and peer-reviewed guideline reviews.